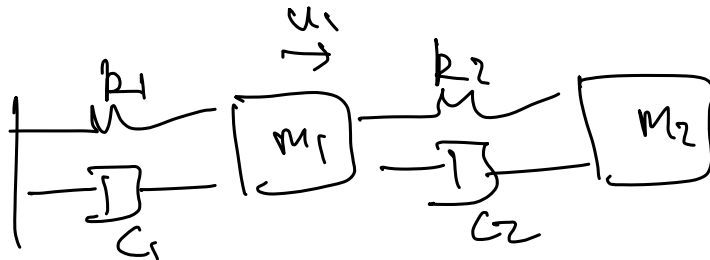


$$x = \begin{bmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \end{bmatrix} \quad \text{input } u \rightarrow m_1$$



$$m_1 \ddot{x}_1 + c_1 \dot{x}_1 + k_1 x_1 + c_2 (\dot{x}_1 - \dot{x}_2) + k_2 (x_1 - x_2) = u$$

$u = \text{feedforward} + \text{PID}$

$$u_{\text{PID}} = K_p e + K_i \int e dt + K_d \dot{e}$$

$$e = r - x_1, \quad \dot{e} = \dot{r} - \dot{x}_1$$

$$u_{\text{ff}} = m_1 \ddot{r} + c_1 \dot{r} + k_1 r + c_2 (\dot{r} - \dot{x}_2) + k_2 (r - x_2)$$

$$u = \underline{\underline{u_{\text{ff}} + u_{\text{PID}}}}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1 + k_2}{m_1} & -\frac{c_1 + c_2}{m_1} & \frac{k_2}{m_1} & \frac{c_2}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & \frac{c_2}{m_2} & -\frac{k_2}{m_2} & -\frac{c_2}{m_2} \end{bmatrix}$$

$$r = 0.8 \sin(2\pi t)$$

$$B = \begin{bmatrix} 0 \\ 1/m_1 \\ 0 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Min-order observer

$$\dot{\tilde{\eta}} = \underbrace{\begin{bmatrix} A_{bb} - \hat{K}_e A_{ab} \end{bmatrix}}_{\hat{A}} \tilde{\eta} + \underbrace{\begin{bmatrix} \hat{A} \hat{K}_e + A_{ba} - \hat{K}_e A_{aa} \end{bmatrix}}_{\hat{B}} y + \underbrace{\begin{bmatrix} B_b - \hat{K}_e B_a \end{bmatrix}}_{\hat{F}} u$$

$$\begin{bmatrix} \dot{x}_a \\ \dot{x}_b \end{bmatrix} = \underbrace{\begin{bmatrix} A_{aa} & A_{ab} \\ A_{ba} & A_{bb} \end{bmatrix}}_A + \underbrace{\begin{bmatrix} B_a \\ B_b \end{bmatrix}}_B u$$

x A B

$$\tilde{x} = \underbrace{\begin{bmatrix} 0 \\ I_{n-1} \end{bmatrix}}_{\hat{C}} [\tilde{x}_b - \hat{K}_e y] + \underbrace{\begin{bmatrix} 1 \\ -\hat{K}_e \end{bmatrix}}_{\hat{D}} y$$

$$\tilde{x} = \hat{C} \tilde{\eta} + \hat{D} y$$

